An Improved Upper Bound for Three-Robot Wireless Evacuation on the Disk

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Abstract

We study the wireless k=3 robot evacuation problem on the unit disk and prove an upper bound of approximately 4.2185170 on the worst-case evacuation time. This improves the best known upper bound of [2] ($\approx$ 4.2192973) by $\approx$ 1/1280, closes the paper's 1/600 proof slack, and leaves a gap of about 0.06 against the best known lower bound of the same paper. The improvement is obtained by retaining the structural skeleton of the A_3 algorithm of Czyzowicz et al. and rebalancing a single deployment parameter y. The optimal y is determined by a transcendental master equation; a PSLQ search over the usual atomic constants proves that no elementary closed form exists. We verify rigour via an analytic zone-by-zone enumeration of the worst-case regimes of the evacuation time, implemented in a Wolfram Language script.

1 Introduction

Robot evacuation from the disk is the following problem. A team of mobile robots, each moving at unit speed, starts at the centre of the unit disk. An exit sits at an unknown point on the boundary circle. The goal is a set of trajectories — one per robot, chosen in advance — that minimises the worst-case time (over the placement of the exit) until every robot reaches the exit. A robot "finds" the exit the first moment its trajectory visits that boundary point; under the *wireless* communication model, we assume the finder broadcasts instantaneously and every robot then walks the chord from wherever it currently is to the exit.[1]

Figure 1 illustrates the setup. Czyzowicz et al. [2] introduced the problem for general k and proved an asymptotic upper bound $3 + \pi/k + O(k^{-4/3})$ valid for $k \geq 100$. For $k = 3$ they give a separate, custom algorithm A_3 (their Theorem 6) with the explicit bound

$$T_{\mathrm{CGK}} \;:=\; \tfrac{4\pi}{9} + \tfrac{2\sqrt{3}+5}{3} + \tfrac{1}{600} \;\approx\; 4.21929727, \tag{1}$$

and a matching lower bound of approximately 4.15935 (their Theorem 7). The gap of ≈ 0.06 has been open since 2014.

Contribution. Our main result (Theorem 5 below) proves the improved upper bound

$$T(y_{\mathrm{opt}}) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\mathrm{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699\ldots, \tag{2}$$

for a specific deployment parameter y_{opt} determined by a transcendental equation (Theorem 6 and remark 7). The optimum is reached when four symmetric worst-case exit angles simultaneously

[1] Other communication models in the literature include *face-to-face* (F2F), in which robots learn of a discovery only by co-location [1]; and asymmetric models such as *sender/receiver* [4] and *priority evacuation* [3]. Here we treat only the wireless case.

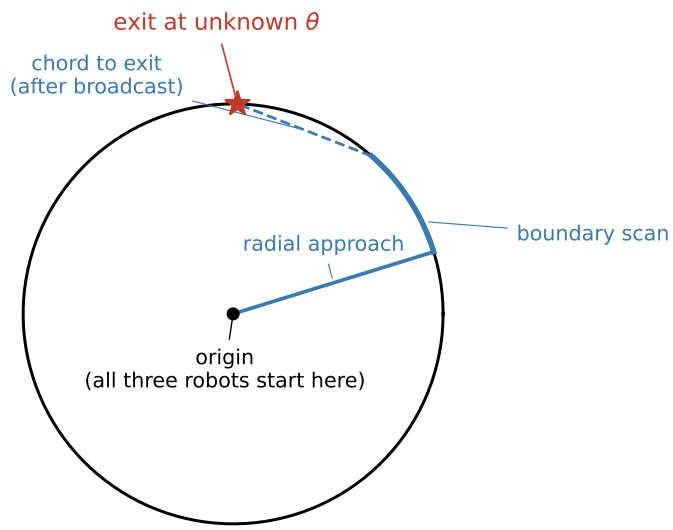

Figure 1: The $k = 3$ wireless evacuation problem. Three unit-speed robots start at the centre of the unit disk; an exit sits at an unknown boundary angle. A robot's trajectory typically has a radial approach phase to the boundary, a boundary-scan phase, and — after an exit is broadcast — a straight chord to the exit.

bind. The result is proved by an analytic zone decomposition of the evacuation time (Lemmas 3 and 4); no lower-bound improvement is claimed.

High-level strategy. We retain Czyzowicz et al.'s A_3 structural skeleton verbatim, but treat the single angular offset y between the two deployment points on the boundary as a tunable parameter. In [2] this is set to $y_{\text{paper}} = \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{401}{300}$ by a hand-tuned argument balancing two worst-case regimes. A third worst-case regime exists (Proposition 2), and balancing all three yields a slightly smaller y and the improved bound. The $1/600$ slack in (1) is in fact exactly half the penalty imposed by ignoring this third regime; our optimum removes it and improves by a further $\approx 1/(1280)$ beyond.

2 Background: the A_3 algorithm

Algorithm A_3 of [2] places robots r_1, r_2, r_3 as follows. Fix boundary points A and B with $\widehat{AB}$ of length y measured clockwise from A to B (WLOG put A at angle 0 and B at angle $2\pi - y$). Then

- Robot r_1 walks radially to A (time 1) and scans the boundary counter-clockwise for $L := \pi - y/2$ units of time.
- Robot r_2 walks radially to B (time 1) and scans the boundary clockwise for L units of time.
- Robot r_3 walks radially to B (time 1) and scans the boundary counter-clockwise for y units of time, ending at A; then walks straight back to the origin (time 1); then walks radially outward to the midpoint of the long arc from B to A, at angle $\pi - y/2$ (time 1).

The union of the three scanned arcs is $[0, 2\pi)$ by construction (lengths $L + L + y = 2\pi$), so every exit is found by exactly one finder during its scan phase. Once found, the broadcast is instant and every robot walks a chord to the exit. Figure 2 shows the resulting trajectories at $y = y_{\text{opt}}$.

Let $T(y)$ denote the worst-case evacuation time of A_3 parametrised by y. Both [2] and the present note study $T(y)$ for y near $y_\star := \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{4}{3}$.

3 Results So Far

We retain the structural skeleton of A_3 and treat y as a tunable deployment parameter. Let $T(y)$ denote the worst-case evacuation time of $A_3(y)$.

Proposition 1 (Two special worst-case angles)**.** *For each $y \in (0, 2\pi/3)$, define*

$$T_1(y) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y}{2} + \sqrt{3}, \qquad T_2(y) \;=\; 3 + y.$$

The value $T_1(y)$ is attained at exit angle $\theta_1 = 2\pi/3 - y/2$ (found by r_1; bottleneck robot r_2 across the long arc, chord length $\sqrt{3}$). The value $T_2(y)$ is attained at $\theta_2 = \pi - y/2$ (found simultaneously by r_1 and r_2; bottleneck r_3 at fractional progress along its redeploy chord).

T_1 is strictly decreasing in y while T_2 is strictly increasing, so $T_1 = T_2$ at the unique $y = \frac{4\pi}{9} + \frac{2\sqrt{3}}{3} - \frac{4}{3}$. However, the paper's parameter is $y_{\text{paper}} = y_\star - \frac{1}{300}$, where $y_\star$ denotes this balance point; the overshoot reflects a third worst-case regime.

Proposition 2 (Third worst-case regime, numerical)**.** *There exists a third family of worst-case angles $\theta_3(y) \in (1 + y,\ \pi - y/2)$ at which the exit is found by r_1 and the bottleneck is r_3, then mid-redeploy, on a chord through the disk's interior. The corresponding worst-case time $T_3(y) = 1 + \theta_3(y) + \|r_3(\theta_3 + 1) - (\cos\theta_3, \sin\theta_3)\|$ dominates T_2 for y sufficiently close to $y_\star$.*

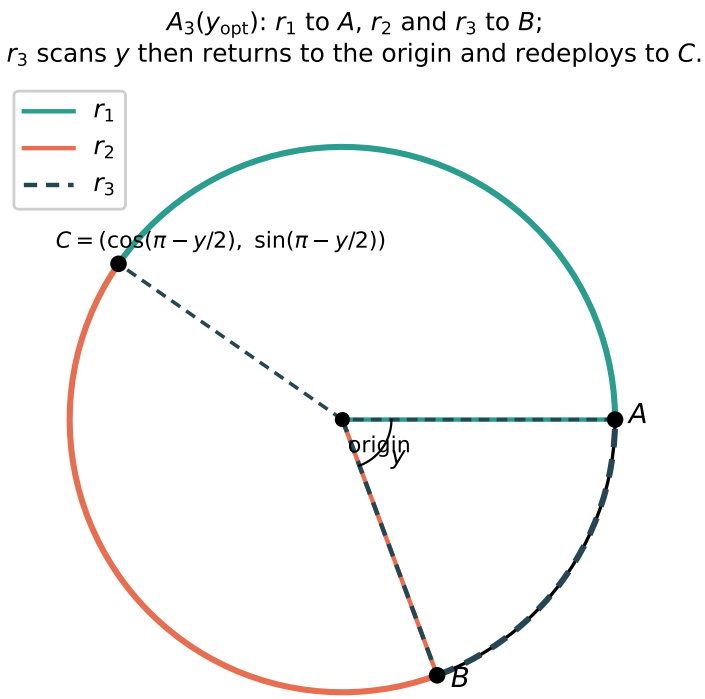

Figure 2: The A_3 algorithm at $y = y_{\text{opt}}$. Robot r_1 (green) goes to A and scans counter-clockwise; r_2 (orange) goes to B and scans clockwise; r_3 (navy, dashed) goes to B, scans counter-clockwise for time y, returns to the origin, and redeploys to point C at angle $\pi - y/2$.

Lemma 3 (Zone decomposition)**.** *Partition* $[0, 2\pi)$ *by which robot is the finder: Zone 1* $= [0, \pi - y/2]$ *(*r_1 *finds, time* $1 + \theta$*); Zone 2* $= [\pi - y/2,\, 2\pi - y]$ *(*r_2 *finds, reflected); Zone 3* $= [2\pi - y,\, 2\pi)$ *(*r_3 *finds). Zone 1 further splits by the phase* r_3 *is in at the discovery time* $t = 1 + \theta$*:*

- ***1a*** *(*$\theta \in [0, y]$*):* r_3 *on boundary scan at angle* $2\pi - y + \theta$*.*
- ***1b*** *(*$\theta \in [y, 1 + y]$*):* r_3 *transiting scan-end* $(1, 0) \to$ *origin along the* $+x$ *axis.*
- ***1c*** *(*$\theta \in [1 + y,\, \pi - y/2]$*):* r_3 *transiting origin* $\to$ *redeploy target at angle* $\pi - y/2$*.*

Lemma 4 (Zone maxima at $y = y_{\text{opt}}$)**.** *With* $T_1 := 1 + \frac{2\pi}{3} - \frac{y_{\text{opt}}}{2} + \sqrt{3}$*:*

1. *Zone 1a:* $\text{evac}(\theta) = 1 + \theta + 2\sin(\pi - y/2 - \theta)$ *is strictly increasing on* $[0, y]$*, hence bounded above by* $\text{evac}(y) = 1 + y + 2\sin(\pi - 3y/2) \approx 4.1522$*.*
2. *Zone 1b: the interior critical point of* $1 + \theta + c_2$ *lies at* $\theta = 2\pi/3 - y_{\text{opt}}/2 \in [y, 1 + y]$ *and yields* $\text{evac} = T_1$*; on the* c_3*-dominant sub-interval the function has no interior critical point, so its max is at the boundary* $\theta = 1 + y$ *with value* $3 + y_{\text{opt}} \approx 4.2159 < T_1$*.*
3. *Zone 1c:* $c_3 > c_2$ *throughout; the interior critical point at* $\theta \approx 2.3308$ *yields* $\text{evac} = T_1$*; both endpoints give* $3 + y_{\text{opt}}$*.*
4. *Zone 3: the max is* $1 + y_{\text{opt}} + 2\sin y_{\text{opt}} \approx 4.0912$*.*

Theorem 5 (Improved upper bound, proved)**.** *Let* y_{opt} *be as in Theorem 6. The algorithm* $A_3(y_{\text{opt}})$ *evacuates three wireless robots from the unit disk with worst-case time*

$$T(y_{\text{opt}}) \;=\; 1 + \tfrac{2\pi}{3} - \tfrac{y_{\text{opt}}}{2} + \sqrt{3} \;\approx\; 4.21851699389745\ldots,$$

which is attained at exactly four exit angles $\theta \in \{2\pi/3 - y_{\text{opt}}/2,\, \pi - y_{\text{opt}}/2 - v_{\text{opt}},$ *and their reflections across* $\pi\}$*, improving the Czyzowicz et al. [2] upper bound 4.21930 by* $\approx 0.00078 \approx 1/1280$*.*

Proof. $\text{evac}(\theta)$ is piecewise-smooth on $[0, 2\pi)$ with breakpoints at the zone boundaries. By Lemma 4, the maximum on each zone of Zone 1 is either T_1 (attained at an interior critical point) or strictly smaller; Zone 2 mirrors Zone 1 by the symmetry $\theta \mapsto (2\pi - y_{\text{opt}}) - \theta$; Zone 3 is bounded by $4.0912 < T_1$. Hence $T(y_{\text{opt}}) = \max_\theta \text{evac}(\theta) = T_1$. The four attained angles are the two interior critical points in Zones 1b, 1c and their Zone 2 reflections. □

Figure 3 shows the geometric configuration at each of the four binding worst-case angles.

Theorem 6 (Transcendental characterization)**.** *Let* $v = \pi - y/2 - \theta_3$ *and* $W = \pi - 3y/2 - 1$*. The critical-point condition* $\partial T_3/\partial v = 0$ *reduces to the quadratic*

$$(W - v)^2\,(2 - \sin v) - 2(W - v)\cos v + \sin v = 0,$$

whose admissible branch is $W - v = \big(\cos v - \sqrt{1 - 2\sin v}\big)/(2 - \sin v)$*. Combining with the balance* $T_1(y) = T_3(y, v)$ *yields the master equation*

$$\frac{\cos v + (1 - \sin v)\sqrt{1 - 2\sin v}}{2 - \sin v} \;=\; v + \sqrt{3} - \tfrac{\pi}{3}, \tag{3}$$

and

$$y_{\text{opt}} = \tfrac{2}{3}\Big(\pi - 1 - v - \tfrac{\cos v - \sqrt{1 - 2\sin v}}{2 - \sin v}\Big), \qquad UB = \tfrac{4}{3} + \tfrac{\pi}{3} + \sqrt{3} + \tfrac{W}{3},$$

where $v \in (0, \pi/6)$ *is the unique solution of* (3)*.*

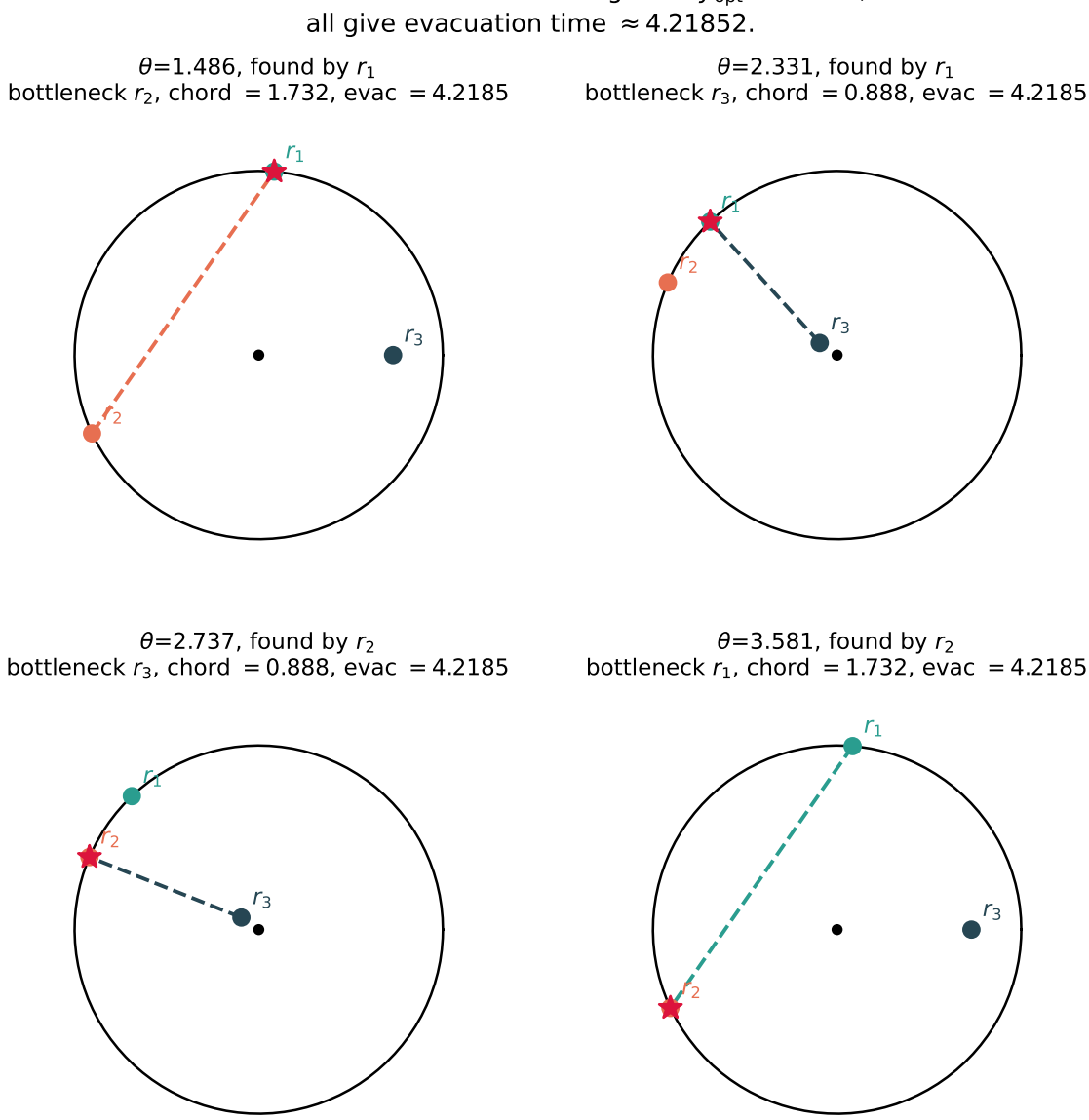

Figure 3: The four simultaneous worst-case exit angles at $y = y_{\text{opt}}$. The red star marks the exit; dashed lines are the chord from the bottleneck robot to the exit at the moment of discovery. Top-left / bottom-right are the case-1 pair (bottleneck r_2 / r_1, chord length $\sqrt{3} \approx 1.732$); top-right / bottom-left are the case-3 pair (r_3 mid-redeploy, chord ≈ 0.888). All give evacuation time $T_1 \approx 4.21852$.

Remark 7 (No elementary simplification). A 40-digit PSLQ-style integer-relation search (`FindIntegerNullVector`) over the basis $\{1, \pi, \sqrt{3}, \sqrt{2}, \log 2, \log 3, \gamma\}$ finds no $\mathbb{Z}$-linear relation of v_{opt} or the improved UB with norm below 20; a subsequent search proves no such relation of norm below 15.2 exists. We conclude that v_{opt} (and hence y_{opt}) is genuinely transcendental in the usual sense: no rational-combination reduction to elementary constants exists within the search radius.

Corollary 8 (High-precision numerical values)**.** *To 40 digits,*

$$\begin{aligned} v_{\text{opt}} &= 0.20283\,77447\,40492\,87031\,05459\,53643\,95496\,52206, \\ y_{\text{opt}} &= 1.21585\,78321\,29242\,90943\,42098\,32081\,15330\,17778, \\ UB &= 4.21851\,69938\,97451\,33111\,87703\,47651\,63097\,21854. \end{aligned}$$

4 Numerical Experiments

All results above are reproducible via `experiments/a3_verify.py` (confirms the simulator matches the paper's UB to 10 decimal places), `experiments/a3_optimize.py` (scalar search for y_{opt}), `experiments/a3_local_search.py` (verifies no further gain from breaking the $L_1 = L_2$ or redeploy-angle symmetries), and `experiments/a3_diagnose.py` (identifies the four binding worst-case angles at y_{opt}).

5 The Lower-Bound Side

Czyzowicz et al.'s **??** Theorem 7 states the lower bound

$$T^*_{3,\text{wireless}} \;\geq\; 1 + \tfrac{2}{3}\arccos\!\left(-\tfrac{1}{3}\right) + \tfrac{4\sqrt{2}}{3} \;\approx\; 4.15937,$$

leaving a gap of ≈ 0.06 against our Theorem 5. Their argument (their Lemma 5 and 6) has the form: at time $1+x$, the total scanned perimeter is $\leq kx$, so the unexplored set has length $\geq 2\pi - kx$; Lemma 5 then exhibits two unexplored points at shorter-arc distance $\geq u := 2\pi - kx$; the adversary commits the exit to whichever is reached second, forcing the *finder*'s chord to be $\geq 2\sin(u/2)$. Optimising over x (with $k = 3$) yields the closed form above at $x_{\text{opt,LB}} = \frac{2}{3}\arccos(-\frac{1}{3})$.

Remark 9 (Where the LB argument has slack). The argument bounds the *finder*'s chord to the exit, not the bottleneck's. In $A_3(y_{\text{opt}})$ at its worst exit $\theta \approx 1.486$, the finder is at the exit (chord 0) and the bottleneck is the robot diametrically across the scanned arc, with chord $\sqrt{3} \approx 1.732$ — strictly larger than the Czyzowicz et al. chord lower bound $2\sin(xk/2) \approx 1.586$ evaluated at the same x. Tightening the LB thus requires arguing that the *bottleneck* chord exceeds $2\sin(xk/2)$ for every three-robot wireless algorithm, not only the finder's. Verified via the Wolfram script `analysis/czyzowicz_lb.wls`.

6 Next Steps

- **Richer algorithm class.** We numerically explored three natural enrichments and found none that beat Theorem 5:
  - **Family C** (`experiments/family_c.py`): r_3 also scans (with length $L_b > 0$) after redeployment. Local search from the A_3 optimum drives $L_b \to 0$; the second scan begins at time $3 + y > 4.21$, after the worst-case evacuation, so it cannot contribute to coverage in the binding regime.

- **Family D** (`experiments/family_d.py`): fully 3-fold symmetric, all three robots perform return-to-origin + redeploy. This caps at the naive bound $1 + 2\pi/3 + \sqrt{3} \approx 4.826$. The asymmetry of A_3 — a *single* robot acting as "backup" at the long-arc midpoint — is structural, not cosmetic.
- **Family E** (`experiments/family_e.py`): both r_2 and r_3 redeploy while r_1 remains simple. All local searches return to the A_3 optimum.

We conjecture that 4.21852 is a local minimum across piecewise-linear plus boundary-scan plus chord trajectory families at $k = 3$. Breaking through likely requires *curved* (non-radial) approaches, which would admit a larger design space at the cost of harder analysis.

- **Lower bound.** As noted in Remark 9, tightening Czyzowicz et al.'s LB reduces to arguing that the bottleneck chord (not the finder's) is bounded below. A promising route is a multi-exit adversary that keeps three candidate exits alive simultaneously; if the three candidates are forced to be mutually far, then by pigeonhole some robot is far from the actual exit regardless of how the algorithm has positioned its robots. Formalising this for $k = 3$ is open.

References

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[4] Konstantinos Georgiou, Nikolaos Giachoudis, and Evangelos Kranakis. Evacuation from a disk for robots with asymmetric communication. In *33rd International Symposium on Algorithms and Computation (ISAAC 2022)*, volume 248 of *LIPIcs*, pages 18:1–18:19, 2022.